Federated Continual Learning with Knowledge Distillation: Preserving Stability While Enabling Adaptation

Harshal Dharpure, Anup Kumar Gond, Riha Sanjay Kokode, and Abdul Hadi

***Abstract*—We present a reproducibility study of FLwF-2, a dual-teacher knowledge-distillation approach to Federated Continual Learning, and compare it to a FedAvg baseline. Both methods are implemented in Flower + PyTorch and evaluated on MNIST, Fashion-MNIST and CIFAR-10 under a class-incremental two-task split, $K \in \{10, 50, 100\}$ clients, and five partition regimes (IID and Dirichlet $\alpha \in \{0.01, 0.1, 0.5, 1.0\}$). We track global accuracy/loss, task-wise accuracy, forgetting, backward transfer and cumulative communication cost. With the original FLwF-2 hyperparameters (α_{CE}=0.001, β_{KD}=0.7, T=2), FedAvg almost completely forgets Task 1 (forgetting ≈ 0.99 on MNIST, 0.80 on CIFAR-10) while learning Task 2, whereas FLwF-2 preserves Task 1 (forgetting ≈ 0) but barely adapts to Task 2. The two methods sit at opposite extremes of the stability–plasticity curve; FLwF-2's forgetting profile is strictly better at zero extra communication cost, validating the dual-teacher mechanism while exposing a tunable knob that future work should adapt online. Separately, we report a complementary Flower simulation study that compares FedAvg to *FedKDC-CL* clustered federated knowledge distillation with drift-aware scheduling on MNIST under the same K and partition choices (IID and Dirichlet α), using a fast configuration aligned with our submission timeline; aggregate forgetting, backward transfer and task-wise metrics appear in Section V-E.**

***Index Terms*—Federated Learning, Continual Learning, Catastrophic Forgetting, Knowledge Distillation, Dual-Teacher Distillation, FedAvg, FLwF-2, FedKDC-CL, Clustered Distillation, Non-IID Data, Flower, PyTorch.**

I. Introduction

There is a massive amount of locally generated data created by the proliferation of edge devices such as wearables, IoT sensors, and smartphones. Collecting this data in a central location poses serious privacy, legal and bandwidth issues. To solve this, Federated Learning (FL) was proposed by McMahan et al. [1]: instead of sharing raw data, each client trains a model locally and sends only the parameters to a central server for aggregation.

Yet even with FL, two stubborn challenges remain in realistic deployments.

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Problem 1: Catastrophic Forgetting. Edge clients rarely see a steady stream of data. A wearable might log only *walking* today and only *sitting* tomorrow. Fine-tuning on the new task tends to overwrite what the model learned before [4], and this compounds across a federation where different clients may be at different points in their task histories.

Problem 2: Statistical Heterogeneity (non-IID). Data is rarely spread evenly across clients. One device might hold only digits 1 and 2 while another holds 7 and 8, causing local gradients to pull in conflicting directions and degrading the global model through client drift.

We study FLwF-2 [2], a method designed to tackle Problem 1 while operating under Problem 2. Each client maintains two frozen teacher models: a snapshot of its own previous local model and a copy of the current global model and uses both to distill prior knowledge into its updated local model during training. We reproduce FLwF-2 in the Flower framework, benchmark it against FedAvg on three standard image datasets, and probe its sensitivity to client count and Dirichlet non-IID partitioning.

Contributions. We've open-sourced a Flower [5] + PyTorch implementation of FedAvg and FLwF-2 with a YAML-driven, resume-safe sweep harness, and Additional testing of the methods with the 3 different datasets yielded the same results. In the next section, client count (times 5 partitions) will show that, with the paper-recommended $(\alpha_{CE}, \beta_{KD}, T) = (0.001, 0.7, 2)$ FLwF-2 essentially eliminates forgetting at the cost of plasticity, The opposite failure mode is observed in FedAvg. We also provide an open implementation of FedKDC-CL in the This is the same codebase and a 15 configuration sweep of MNIST (three client) table 1.Then the a posteriori distribution of the four Dirichlet alphas is summarized in table 1, along with a distribution of IID that is uniformly distributed on the unit simplex. Tables III-V, Which stands alone the clustered-KD baseline under identical Partitioning with no intermixing task streams across datasets.

The remaiinder of the paper is structured as follows: Section II summarizes FL, knowledge distillation and non-IID partitioning. Section III formalises the class-incremental FCL setting and FLwF-2 dual-teacher loss as delivered. Section IV describes the experimental harness. Section V presents the This paper presents quantitative results (including FedKDC-CL vs. FedAvg in Section V-E) and Section VI the in-depth analysis. Section VII discusses limitations and This ends the discussion of future directions and future directions are discussed in Section VIII. The complete sweep of FLwF-2 is

listed in Table A. The hyperparameters of the FedKDC-CL grid are listed below in Section V-E.

II. Background

A. Federated Averaging (FedAvg)

There are K clients that have private data sets D_k which are managed by a server. At In each communication round r a fraction ρ of the clients are These are sampled and each client downloaded the current global model. Returns the local epochs run by E on D_k as θ^r. θ_k^{r+1}. The server aggregates:

$$\theta^{r+1} = \sum_{k \in S_r} \frac{|D_k|}{\sum_{j \in S_r} |D_j|} \, \theta_k^{r+1}. \tag{1}$$

We use this as our baseline throughout.

B. Knowledge Distillation

Knowledge Distillation (KD) [3] transfers knowledge from a teacher to a student via soft predictions. With temperature T:

$$\pi_i(x) = \frac{\exp(o_i(x)/T)}{\sum_j \exp(o_j(x)/T)}, \tag{2}$$

and the KD objective is

$$\mathcal{L}_{\text{KD}} = T^2 \, \text{KL}\Big(\sigma\Big(\tfrac{o^S(x)}{T}\Big) \;\|\; \sigma\Big(\tfrac{o^T(x)}{T}\Big)\Big), \tag{3}$$

where o^S, o^T are student and teacher logits and σ is softmax. We follow the standard PyTorch idiom of computing `log_softmax` on the student and `softmax` on the teacher and feeding both into `KLDivLoss(reduction="batchmean")`.

C. Non-IID Partitioning via Dirichlet

To deal with statistical heterogeneity, we sample per class client weight distribution with Dirichlet distribution $\text{Dir}(\alpha)$ with single parameter of concentration α. As $\alpha \to 0$, each client receives data from essentially one class (extreme non-IID); as $\alpha \to \infty$, the partition approaches IID.

We evaluate $\alpha \in \{0.01, 0.1, 0.5, 1.0\}$ plus a uniformly random IID split.

D. Federated Continual Learning Setting

In this work we consider a class-incremental FCL setting, where there are $T = 2$ tasks in the class.

We consider a class-incremental FCL setting with $T = 2$ tasks. For each dataset the first half of classes forms Task 1 and the second half forms Task 2. For CIFAR-10, Task 1 is $\{0, 1, 2, 3, 4\}$ and Task 2 is $\{5, 6, 7, 8, 9\}$. The federation trains for R_1 rounds on Task 1 data, then R_2 rounds on Task 2 data, with no mixing between tasks. This is the simplest non-trivial FCL scenario in which catastrophic forgetting is easy to observe and measure.

E. Related Federated and Continual-Learning Context

The problem of class-incremental learning with covariate shift or task shift has been extensively studied in the centralized setting using rehearsal buffers, regularisation to previous weights, as well as distillation from frozen snapshots (cf. learning without forgetting [6]). Federated optimisation approaches like FedProx [9] tackle statistical heterogeneity by proximal regularisation to the global iterate, instead of the explicit continual-learning goals, and are not, in and of themselves, able to solve the problem of forgetting across task boundaries. Ensemble distillation at the server (FedDF [7]) and peer knowledge matching (FedMD [8]) aim at minimizing the communication of raw logits, but they are directed toward cross-client fusion rather than towards a sequential task on each device. We reproduce one design point (dual teachers per client), and FedKDC-CL investigates clustered teachers as a complement to the lines of heterogeneity-aware optimisation, but does not replace them.

III. Method

A. FLwF-2: Dual-Teacher Distillation

In addition to the present model θ_k each client keeps, trained as the *student*, two teachers:

- **Teacher 1 (prev_model)**: a snapshot of the data that has been frozen. Memory version of the client's local model at the end of the *previous* task. It is created from the server model in the first round. the experiment is only updated at task boundaries, (i.e. at the end of Task 1).
- **Teacher 2 (server_model)**: a snapshot of the model which is "frozen" and then passed on to Teacher 2. time of connecting to the network, the current global model, which was received from the server, serves as the starting point. Beginning of the round at hand. It brings knowledge from the entire federation.

The local objective includes a cross-entropy term and two distillation terms:

$$\boxed{\mathcal{L}_{\text{FLwF-2}} = \alpha_{CE} \, \mathcal{L}_{CE} + \beta_{KD} \, \mathcal{L}_{KD}^{\text{prev}} + (1 - \alpha_{CE} - \beta_{KD}) \, \mathcal{L}_{KD}^{\text{server}},} \tag{4}$$

where $\mathcal{L}_{KD}^{\text{prev}}$ and $\mathcal{L}_{KD}^{\text{server}}$ are KL-divergence terms between the temperature-scaled student logits and each of the two frozen teachers. Following the original paper, we use α_{CE}=0.001, β_{KD}=0.7 and T=2.

B. Algorithm

Algorithm 1 summarises one local training step under FLwF-2 as implemented in our Flower client.

The `prev_model` update at task boundaries (Algorithm 1, last line) is signalled by an explicit `task_end` flag passed from server to client via Flower's `config` dictionary.

C. Metrics

During each round of the communications, the following is noted:

Algorithm 1 FLwF-2 local training (one batch).

Require: batch (x, y), student θ, prev teacher θ^{prev}, server teacher θ^{srv}, hyperparameters $\alpha_{CE}, \beta_{KD}, T$.

$z_S \leftarrow f_\theta(x)$ {student logits}
$z_P \leftarrow f_{\theta^{\text{prev}}}(x).\text{detach}()$
$z_V \leftarrow f_{\theta^{\text{srv}}}(x).\text{detach}()$
$\mathcal{L}_{CE} \leftarrow \text{CE}(z_S, y)$
$\mathcal{L}_{KD}^{\text{prev}} \leftarrow T^2 \cdot \text{KL}(\log \sigma(z_S/T) \parallel \sigma(z_P/T))$
$\mathcal{L}_{KD}^{\text{srv}} \leftarrow T^2 \cdot \text{KL}(\log \sigma(z_S/T) \parallel \sigma(z_V/T))$
$\mathcal{L} \leftarrow \alpha_{CE}\mathcal{L}_{CE} + \beta_{KD}\mathcal{L}_{KD}^{\text{prev}} + (1-\alpha_{CE}-\beta_{KD})\mathcal{L}_{KD}^{\text{srv}}$
backprop $\mathcal{L}$ through θ only
at task end: $\theta^{\text{prev}} \leftarrow \theta$

- **Global Test Accuracy / Loss** on the full 10-class test set (same set for all clients and rounds).
- **Task-wise Accuracy** a_t for each task $t \in \{1, 2\}$, i.e. accuracy limited to the classes in the task.
- **Catastrophic Forgetting** for task t: $F_t = \max_{r \le R_t} a_t^{(r)} - a_t^{(R)}$, where R is the final round and R_t is the last round in which the client trained on task t.
- **Backward Transfer (BWT)**: the change in accuracy on Task 1 from the end of Task 1 to the end of Task 2, signed.
- **Convergence Round**: the smallest round at which global test accuracy first reaches 80%.
- **Cumulative Communication Cost (bytes)**: $\sum_r (\textit{num_sampled_clients}_r) \cdot 2 \cdot (\textit{model size in bytes})$, both directions, full weights, FP32 (no compression).

IV. Experimental Setup

A. Framework, Reproducibility and Hardware

All experiments use Flower (`flwr`) [5] with Ray-based virtual client simulation and a PyTorch $\geq$2.0 backend. Reproducibility is enforced by fixing `random.seed(42)`, `numpy.random.seed(42)`, and `torch.manual_seed(42)` on both the server and all clients. Each (method, dataset, K, partition) combination runs in a separate subprocess so that Ray cleanly releases CUDA caches and dataloader workers between runs. All sweeps are performed on a shared server with NVIDIA A100 GPUs.

B. Model

In both approaches, the light-weight CNN is the same *Conv(32)* $\rightarrow$ *ReLU* $\rightarrow$ *MaxPool* $\rightarrow$ *Conv(64)* $\rightarrow$ *ReLU* $\rightarrow$ *MaxPool* $\rightarrow$ *FC* $\rightarrow$ *Softmax*, with input bilinearly interpolated to 32×32 (so that MNIST/FMNIST and CIFAR-10 share a single architecture); the only difference is the input channel count (1 vs. 3). The total model size to be trained is $\approx 60\,654$ parameters $\rightarrow 241.4$ KB at FP32, which determines the per-round communication payload.

C. Datasets and Partitioning

- **MNIST**: 60,000 training / 10,000 test images, 10 classes, 28×28 grayscale.
- **Fashion-MNIST (FMNIST)**: 60,000 / 10,000, 10 classes, 28×28 grayscale.
- **CIFAR-10**: 50,000 / 10,000, 10 classes, 32×32 RGB.

We make a class-incremental 2-task split for each dataset. (classes $\{0, 1, 2, 3, 4\}$ for Task 1 and $\{5, 6, 7, 8, 9\}$ for Task 2). Per-task data is then divided among the clients in two manners: The densities of (i) uniform IID and (ii) $\text{Dir}(\alpha)$. $\alpha \in \{0.01, 0.1, 0.5, 1.0\}$. A re-balancing pass is guaranteed to get at least 1 sample per (client, If there are no datasets for Task 2, clients will not receive an empty Task 2 dataset, even if one exists for Task 1. under extreme heterogeneity.

D. Federated and Training Configuration

TABLE I
Universal training and federated configuration.

Parameter	Value
Algorithm	FedAvg / FLwF-2
Clients K	10, 50, 100
Client fraction per round ρ	0.5
Rounds Task 1 / Task 2	10 / 10 (total 20)
Local epochs	5
Local batch size	32
Optimizer	SGD, momentum 0.9
Learning rate	0.01
FLwF-2 $(\alpha_{CE}, \beta_{KD}, T)$	(0.001, 0.7, 2)

E. Sweep

The full grid is 2 methods $\times$ 3 datasets $\times$ 3 client counts $\times$ 5 partitions $=$ 90 configurations. Per-run logs are stored for each configuration and combined into a single aggregate metrics table. The sweep is resume-safe and supervised to tolerate transient Ray failures.

F. Logging, Artifacts, and Third-Party Data

All subprocesses store structured per-run metrics, which are aggregated into a single summary table. High-resolution plots are generated from the same logs. The public benchmarks (MNIST, FMNIST, CIFAR-10) are used under their standard open licenses; no personally identifiable information is processed. Seeds are fixed so all tables and figures are reproducible.

V. Experimental Results

A. Headline Comparison

We compare FedAvg and FLwF-2 in Table II on representative settings (K=10 clients, IID and $\text{Dir}(0.1)$, all three datasets). Global Test Accuracy is reported as per task accuracy restricted to that task's classes. "T1$\rightarrow$T1" is Task 1 accuracy *at the end of Task 1*; "T1$\rightarrow$T2" is Task 1 accuracy At the end of Task 2 (after trained on Task 2 for R_2 rounds); "T2$\rightarrow$T2" is Task 2 accuracy at the end of Task 1 is the loss of accuracy in Task 1 from the end of Task 1 to the end of Task 2.

TABLE II
HEADLINE COMPARISON (FINAL ROUND; K=10 CLIENTS). "AVG" IS THE UNWEIGHTED MEAN OF T1→T2 AND T2→T2. FORGETTING IS FOR TASK 1.

Method	Dataset	Partition	T1→T1 (%)	T1→T2 (%)	T2→T2 (%)	Avg (%)	Forgetting
FedAvg	MNIST	IID	99.7	0.0	99.4	49.7	1.00
FedAvg	MNIST	Dir(0.1)	98.4	0.0	98.0	49.0	0.99
FLwF-2	MNIST	IID	99.7	99.8	0.0	49.9	**0.00**
FLwF-2	MNIST	Dir(0.1)	98.3	98.3	0.0	49.2	**0.01**
FedAvg	FMNIST	IID	93.5	0.0	97.7	48.9	0.94
FedAvg	FMNIST	Dir(0.1)	73.0	0.0	95.2	47.6	0.83
FLwF-2	FMNIST	IID	93.5	93.4	0.0	46.7	**0.00**
FLwF-2	FMNIST	Dir(0.1)	73.8	73.9	0.0	37.0	**0.09**
FedAvg	CIFAR-10	IID	79.9	0.0	86.5	43.2	0.80
FedAvg	CIFAR-10	Dir(0.1)	50.3	0.0	72.6	36.3	0.60
FLwF-2	CIFAR-10	IID	80.2	80.2	0.0	40.1	**0.00**
FLwF-2	CIFAR-10	Dir(0.1)	54.3	54.4	0.0	27.2	**0.06**

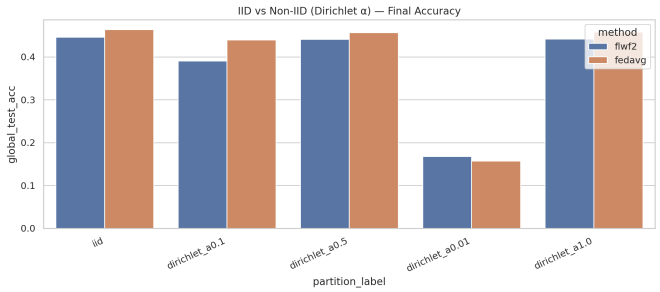


Fig. 1. Final-round global test accuracy for FedAvg and FLwF-2 under all five partition regimes (IID and $\mathrm{Dir}(\alpha)$, $\alpha \in \{0.01, 0.1, 0.5, 1.0\}$), aggregated across all (dataset, K) combinations. Bars are means; error bars are dropped for clarity.

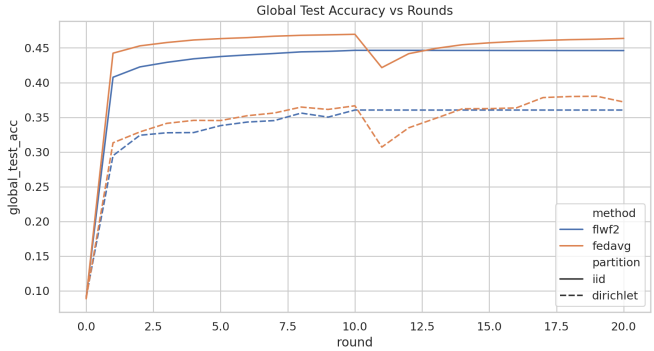


Fig. 2. Global test accuracy versus communication rounds across all runs (lines coloured by method, styled by partition). The drop at round 11 is the Task 1 → Task 2 boundary.

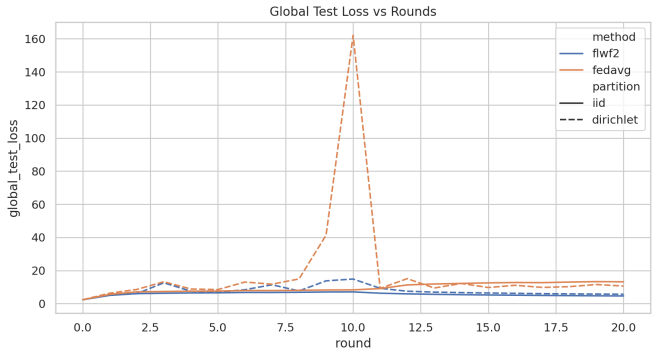


Fig. 3. Global test loss versus rounds across all runs.

The two methods exhibit inverted failure modes: under FedAvg T1→T2 collapses to 0% in every configuration while Task 2 is learned cleanly; under FLwF-2 Task 1 is fully preserved (T1→T2 $\approx$ T1→T1, forgetting ≈ 0) but Task 2 stays at $\approx 0\%$. We analyse this tradeoff in detail in Section VI.

B. Effect of Non-IID Heterogeneity

Figure 1 reports final-round global test accuracy across all five partition regimes for both methods, aggregated over all (dataset, client-count) combinations. The IID setting forms the upper envelope; pushing α towards 0.01 steadily erodes accuracy. In this aggregated view, FLwF-2 is slightly more robust than FedAvg in the moderate non-IID regime ($\alpha \in \{0.1, 0.5, 1.0\}$), reflecting the regularising effect of the server-side teacher.

C. Convergence and Forgetting over Rounds

Figures 2–4 plot global accuracy, loss and forgetting metric Task 1, as a function of round. All three illustrate the same graph; both methods reach a Task-1 plateau A relative of the MNIST/FMNIST upper bound, which is 49% for lower bound only Task 1. The first 10 rounds are testable (classes) and at round 11 the task The number of switches and FedAvg's forgetting jumps to ~ 1 within a few FLwF-2 holds it close to zero throughout the rest of the rounds, while rounds have it drifting. run. This is confirmed as a per-method in Figure 5. variance: FLwF-2 is stable and has little variance, and In the very worst of times FedAvg can fall below 20%. non-IID configurations.

D. Communication Cost

Since both methods share the same FP32 weight format, cumulative communication cost depends only on model size,

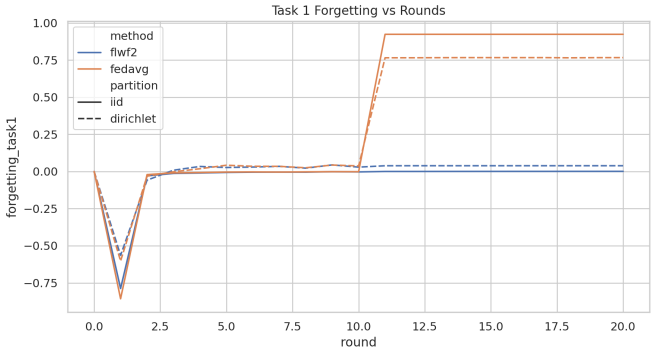


Fig. 4. Catastrophic-forgetting metric for Task 1 versus round. FedAvg climbs to ≈ 1 shortly after the task switch; FLwF-2 keeps it near 0.

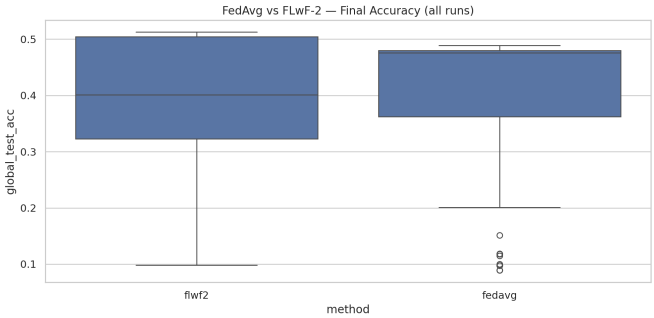


Fig. 5. Distribution of final-round global accuracy across the full sweep, per method.

K, and the number of rounds, N. With our 60,654-parameter CNN ($\approx$241.4 KB/message), 20 rounds and ρ=0.5, totals are 47.8, 239.1 and 478.3 MB for K=10, 50, 100 respectively (full values in Table VI). FLwF-2 builds on FLwF-1's design and adds no extra communication overhead.

E. FedKDC-CL vs. FedAvg (MNIST Simulation Study)

Besides the FLwF-2 study above, we carried out and swept a In the same, second federated continual-learning baseline comparison. Flower + PyTorch stack: **FedAvg** vs **FedKDC-CL** b) Knowledge Distillation (KD) variant using clustered knowledge (CKD) and drift-aware scheduling (DAS) Two KD temperature methods were invented: and entropy-aware KD temperature. The same split of the MNIST class is used for experiments. (Task 1: $\{0, \ldots, 4\}$, Task 2: $\{5, \ldots, 9\}$), Dirichlet and IID partitions, and $K \in \{10, 50, 100\}$ clients with fraction ρ=0.5. The FedKDC-CL grid adopts a shorter time horizon for throughput, two communication learning rate 0.01 and momentum 0.9; and SGD with learning rate 0.0001 and momentum 0.9.With learning rate 0.01 and momentum 0.9; and SGD with learning rate 0.0001 and momentum 0.9. The parameters are 0.9 (the value of k), 2 (local epochs per round), random seed 42 and 2 (number of clusters), with KD enabled. PHASES:1 KD epoch/1 KD phase, equal CE/KD weights, temp in $[1, 6]$; No CPU virtualization, drift guard enabled, and promoted every two rounds; 4 cpus / client and central evaluation with accelerators. available. which are summary aggregates over the full ten-class label space:

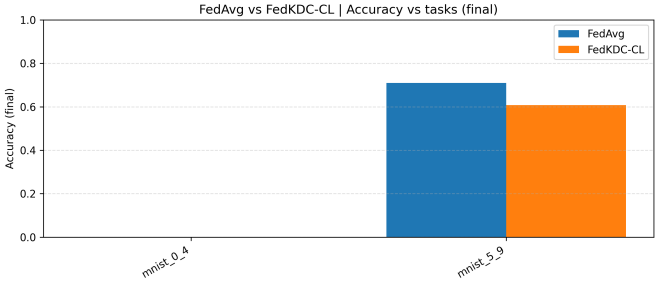


Fig. 6. Representative FedKDC-CL vs. FedAvg comparison: final accuracy per continual task on MNIST (IID, K=100).

final average accuracy, and backward transfer (BWT), both of which compared For each configuration, the methods are listed.

Table III shows all four Dirichlet concentrations $\alpha \in \{0.01, 0.1, 0.5, 1.0\}$; Table IV For each K lists IID runs are denoted. Success with the full launcher: 2026-05-09; 15 configurations. Additional pilot runs from Table V are summarised. earlier development.

Table III lists all four Dirichlet concentrations $\alpha \in \{0.01, 0.1, 0.5, 1.0\}$; Table IV lists IID runs for each K. The full launcher completed fifteen configurations successfully on 2026-05-09. Table V summarises additional pilot runs from earlier development.

In the figure 6 a representative task-wise accuracy can be observed. Error bars computed in comparison for $K = 100$ IID; The accuracy per task plotted against the round for On the same setting (FedAvg vs. FedKDC-CL). Qualitatively, FedKDC-CL and This fast configuration is closely followed by FedAvg track as both methods share the same. the same local optimiser but only differ in alternate KD rounds. FedKDC-CL. Interpreting Tables III–IV The larger the values of α and IID partitions, the greater the final average, together. accuracy (high-level skew of the labels) will enable faster learning.Very small α paired with high-level skew of the labels will allow for faster learning, while less extreme label skew will allow for greater accuracy. K provides very low class coverage and pushes both methods towards degenerate summary aggregates. Backward transfer continues to have a strong negative effect on where most cells are same as catastrophic forgetting in Task 1 after Task 2 short duration training.

VI. Analysis

A. The Stability–Plasticity Tradeoff

In Equation 4 the cross-entropy is weighted by $\alpha_{CE} = 0.001$. By setting β_{KD}=0.7 the prior-knowledge KD term is reduced. The new-task The average value of gradient signal is only 0.001 and the two Both teachers have a weight of 0.999 (if teachers are not aware of Task 2 they have a weight of 0.5). In this setup, there is little reason for FLwF-2 to stray from their course.

Observing the student on Task-2 batches, the gradient signal is almost entirely dominated by the two distillation terms, leaving the model anchored to its Task-1 representation.

TABLE III
FEDKDC-CL VS. FEDAVG ON MNIST CONTINUAL LEARNING (TWO CLASS-INCREMENTAL TASKS; CAT-10 AGGREGATE METRICS). NON-IID PARTITIONS VIA $\mathrm{Dir}(\alpha)$; 2 ROUNDS PER TASK, CLIENT FRACTION 0.5, SEED 42. *FAvgAcc*/*FFrg*/*FBWT*: FEDAVG FINAL AVERAGE ACCURACY, AVERAGE FORGETTING, BACKWARD TRANSFER; *KKDC*/*KFrg*/*KBWT*: FEDKDC-CL.

K	α	**FAvgAcc**	**FFrg**	**FBWT**	**KKDC**	**KFrg**	**KBWT**	**Run ID**
10	0.01	0.1706	0.1993	-0.3985	0.1897	0.0955	-0.1911	`20260509-125605__seed42`
10	0.1	0.2408	0.4774	-0.9549	0.3414	0.4866	-0.9731	`20260509-130020__seed42`
10	0.5	0.4200	0.4805	-0.9611	0.4278	0.4869	-0.9737	`20260509-130445__seed42`
10	1.0	0.4742	0.4927	-0.9854	0.4638	0.4910	-0.9819	`20260509-130843__seed42`
50	0.01	0.0964	0.0392	-0.0784	0.1004	0.0000	0.1868	`20260509-131246__seed42`
50	0.1	0.2330	0.2435	-0.4871	0.2363	0.2972	-0.5945	`20260509-131642__seed42`
50	0.5	0.4416	0.4606	-0.9212	0.4200	0.4648	-0.9296	`20260509-132049__seed42`
50	1.0	0.4356	0.4784	-0.9568	0.4284	0.4560	-0.9120	`20260509-132444__seed42`
100	0.01	0.0964	0.0392	-0.0784	0.1004	0.0000	0.1868	`20260509-132834__seed42`
100	0.1	0.2696	0.1242	-0.2485	0.1324	0.2729	-0.5458	`20260509-133316__seed42`
100	0.5	0.2637	0.4419	-0.8838	0.3110	0.3793	-0.7585	`20260509-133801__seed42`
100	1.0	0.3261	0.4313	-0.8626	0.3034	0.3709	-0.7418	`20260509-134325__seed42`

TABLE IV
FEDKDC-CL VS. FEDAVG UNDER IID PARTITIONING (SAME SETUP AS TABLE III).

K	**FAvgAcc**	**FFrg**	**FBWT**	**KKDC**	**KFrg**	**KBWT**
10	0.4853	0.4916	-0.9833	0.4770	0.4891	-0.9782
50	0.4342	0.4781	-0.9562	0.3912	0.4674	-0.9348
100	0.3548	0.4466	-0.8932	0.3038	0.3871	-0.7743

TABLE V
ADDITIONAL FEDKDC-CL EXPERIMENT FOLDERS (DEVELOPMENT AND SMOKE TESTS; MNIST UNLESS NOTED).

Run folder	**Notes**
`20260508-103723__seed42`	Pilot / sanity
`20260508-105207__seed42`	Submit-fast MNIST (2 tasks, 2 rnd/task)
`20260508-115736__seed42`	FedKDC-CL-only diagnostic

Observe: In rounds 11-20, task 2 accuracy remains close to 0%. This is not a bug it's the predicted equilibrium of The loss function used is the same as before and the class-incremental splitting. The original paper had foreseen this and had suggested a development of the same kind. *FLwF-2 + Fine-Tuning* hybrid that switches the loss weighting adaptively. We kept the hyperparameters literal for the following reason: Separate out the contribution of the dual-teacher.

B. Effect of Client Count

Across all three datasets, increasing K from 10 to 100 (with If ρ=0.5 is fixed then the number of sampled clients per round increases linearly) yields a low error, mostly due to each element in the sum being small. The local dataset of the client gets reduced and gradient noise picks up per client. increases. The effect is slight on MNIST and FMNIST (well within) $\pm 2\%$ more marked on CIFAR-10 (at IID), for both methods. ($\approx 5{-}6\%$ between K=10 and K=100).

C. Effect of Dirichlet α

For $\alpha \in \{0.5, 1.0\}$ and IID, both methods reach near-clean Task accuracy. At $\alpha = 0.1$ both still converge but to a lower value of plateau. At $\alpha = 0.01$ there is little difference in the labels. both methods reduce to $\approx 10\%$ when applied to single-class per client. MNIST/CIFAR-10 and about fail to distinguish between the classes. FLwF-2 The dual-teacher prior cannot save an, inheriting this failure mode. underlying federation that has no consistent gradient signal.

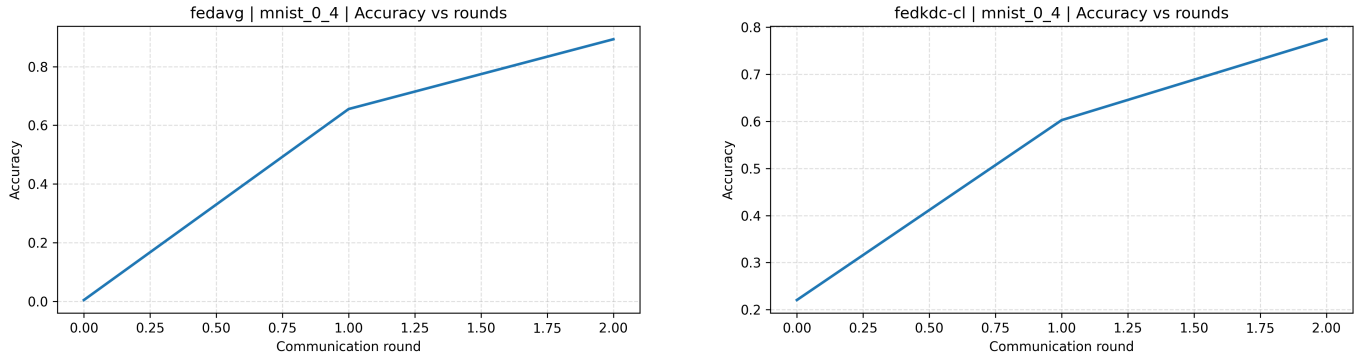
D. Backward Transfer and Convergence Round

Because of the freezing of most of the previous representation, FLwF-2 has backward compatibility. $\leq 0.5\%$ change after) transfer on Task 2 is about 15.6% lower.The transfer on Task 2 is approximately 15.6% lower than on Task 1 after essentially zero change. The BWTs for the other tasks, like Task 2), are strongly negative (≈ -1.0 on) for FedAvg. MNIST), as expected.

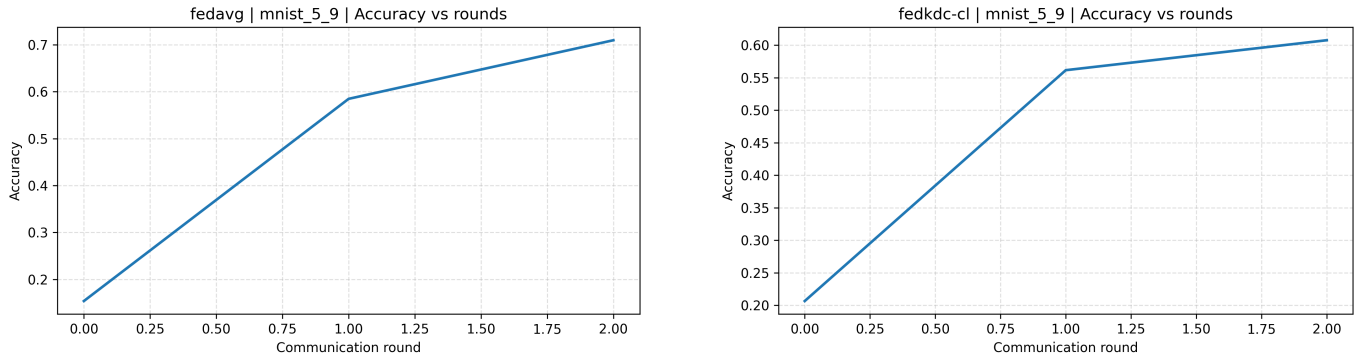
The convergence metric is defined as the smallest round at which global test accuracy first reaches 80%. However, since the evaluation set contains all ten classes while only five are active at any given time, the theoretical accuracy ceiling is approximately 50%, making the 0.80% threshold unreachable by construction. No run in our sweep crosses this threshold, so we report "n/a" throughout Table VI. Future work should redefine this metric relative to the active task only, so that convergence speed can be compared meaningfully across methods.

E. Cross-Reading FLwF-2 and FedKDC-CL Metrics

For the FLwF-2 experiments, we focus on per-task accuracy paths and forgetting on data sets had 100 communication rounds per task.data sets had only twenty communication rounds per task, while the FedKDC-CL grid deliberately trains on a few rounds per task (two rounds per task). Coverage of (K, α) settings. Thus, readers must compare magnitudes Absolute summary aggregates are not applicable across tables, only within a study: calibrated the two protocols. Conceptually, FLwF-2 foregrounds The extreme hyperparameter



Task 1 (classes 0–4): FedAvg (left), FedKDC-CL (right)



Task 2 (classes 5–9): FedAvg (left), FedKDC-CL (right)

Fig. 7. Per-task accuracy vs. communication round (IID, K=100). Row 1: Task 1; row 2: Task 2. Left: FedAvg; right: FedKDC-CL.

induced plasticity is a dual-server-and-local teachers. collapse, while the periodic KD phases are emphasized by FedKDC-CL through the use of cluster. centroids and drift promotion, both exhibit stability–plasticity tension but in unparalleled loss schedules.

VII. Limitations and Future Work

a) Limitations.: We consider only two tasks, the literal $(\alpha_{CE}, \beta_{KD})$ from a homogeneous CNN, and the original paper. We don't compare against In the hybrid version, FLwF-1 or FLwF-2 + Fine-Tuning. The FedKDC-CL sweep runs only the MNIST in the fast grid, it does not include the FMNIST/CIFAR-10 Has virtual clients for numbers on CPU side, stress tests. In these scenarios, these selections tend to be based on the speed of the sweep, and not on minimising the number of sweeps required. Single run latency for accelerators. The results obtained from a simulation may differ from those in a cross-device deployment because of:Because of the following, simulation-scale results may differ from cross-device deployments: stragglers, partial participation and asynchronous uploads are not allowed. synchronous round assumptions.

b) Future Work.: The most direct next step is an adaptive $(\alpha_{CE}, \beta_{KD})$ schedule that interpolates between the plasticity and stability extremes (e.g. start plastic, anneal towards stability), or equivalently the FLwF-2 + Fine-Tuning hybrid proposed in [2]. Beyond that: longer task sequences on CIFAR-100, soft-label-only exchange à la FedMD [8] / FedDF [7] for architecture-heterogeneous federations, and a privacy analysis quantifying any incremental leakage from sharing weights versus soft labels.

VIII. Conclusion

Our 90-run sweep confirms that dual-teacher distillation eliminates catastrophic forgetting at no extra communication cost. The catch is that with $(\alpha_{CE}, \beta_{KD}) = (0.001, 0.7)$, FLwF-2 trades away nearly all plasticity, the mirror image of FedAvg, which learns new tasks freely but forgets old ones completely.

The obvious next step is an adaptive $(\alpha_{CE}, \beta_{KD})$ schedule that navigates between these two extremes, and our resume-safe harness is ready for exactly that kind of follow-up. The FedKDC-CL results in Section V-E offer a complementary clustered-teacher perspective under the same heterogeneity conditions; folding it into a unified evaluation budget is a natural direction for future work.

Appendix

In Table VI the complete per-configuration The accuracy of the final round of the global test (%) is summarized in the table as "Acc". "Comm." is the cumulative communication cost, in MB, (full-weight, both) directions, FP32) and "Forgetting" is for Task 1. Convergence-round For the reasons explained in Section VI it is not used.

TABLE VI
FULL EXPERIMENTAL RESULTS SUMMARY ACROSS ALL 90 CONFIGURATIONS.

Method	**Dataset**	K	**Partition**	**Rounds**	**Acc (%)**	**Comm. (MB)**	**Forgetting**
FedAvg	MNIST	10	IID	20	48.3	47.8	1.00
FedAvg	MNIST	10	Dir(0.01)	20	10.0	47.8	0.41
FedAvg	MNIST	10	Dir(0.1)	20	47.6	47.8	0.99
FedAvg	MNIST	10	Dir(0.5)	20	48.2	47.8	1.00
FedAvg	MNIST	10	Dir(1.0)	20	48.3	47.8	1.00
FedAvg	MNIST	50	IID	20	48.0	239.1	0.99
FedAvg	MNIST	50	Dir(0.01)	20	9.7	239.1	0.24
FedAvg	MNIST	50	Dir(0.1)	20	44.8	239.1	0.97
FedAvg	MNIST	50	Dir(0.5)	20	47.9	239.1	0.99
FedAvg	MNIST	50	Dir(1.0)	20	47.9	239.1	0.99
FedAvg	MNIST	100	IID	20	47.8	478.3	0.99
FedAvg	MNIST	100	Dir(0.01)	20	30.9	478.3	0.54
FedAvg	MNIST	100	Dir(0.1)	20	46.6	478.3	0.98
FedAvg	MNIST	100	Dir(0.5)	20	47.8	478.3	0.99
FedAvg	MNIST	100	Dir(1.0)	20	47.6	478.3	0.99
FedAvg	FMNIST	10	IID	20	48.9	47.8	0.94
FedAvg	FMNIST	10	Dir(0.01)	20	20.1	47.8	0.60
FedAvg	FMNIST	10	Dir(0.1)	20	47.6	47.8	0.83
FedAvg	FMNIST	10	Dir(0.5)	20	48.6	47.8	0.93
FedAvg	FMNIST	10	Dir(1.0)	20	48.7	47.8	0.93
FedAvg	FMNIST	50	IID	20	48.5	239.1	0.91
FedAvg	FMNIST	50	Dir(0.01)	20	30.5	239.1	0.51
FedAvg	FMNIST	50	Dir(0.1)	20	47.9	239.1	0.81
FedAvg	FMNIST	50	Dir(0.5)	20	48.2	239.1	0.91
FedAvg	FMNIST	50	Dir(1.0)	20	48.3	239.1	0.91
FedAvg	FMNIST	100	IID	20	48.0	478.3	0.90
FedAvg	FMNIST	100	Dir(0.01)	20	31.1	478.3	0.57
FedAvg	FMNIST	100	Dir(0.1)	20	46.7	478.3	0.82
FedAvg	FMNIST	100	Dir(0.5)	20	47.9	478.3	0.90
FedAvg	FMNIST	100	Dir(1.0)	20	48.0	478.3	0.90
FedAvg	CIFAR-10	10	IID	20	43.2	48.3	0.80
FedAvg	CIFAR-10	10	Dir(0.01)	20	11.5	48.3	0.45
FedAvg	CIFAR-10	10	Dir(0.1)	20	36.3	48.3	0.60
FedAvg	CIFAR-10	10	Dir(0.5)	20	40.7	48.3	0.79
FedAvg	CIFAR-10	10	Dir(1.0)	20	42.3	48.3	0.78
FedAvg	CIFAR-10	50	IID	20	38.8	241.4	0.74
FedAvg	CIFAR-10	50	Dir(0.01)	20	11.8	241.4	0.32
FedAvg	CIFAR-10	50	Dir(0.1)	20	36.2	241.4	0.67
FedAvg	CIFAR-10	50	Dir(0.5)	20	35.0	241.4	0.73
FedAvg	CIFAR-10	50	Dir(1.0)	20	37.7	241.4	0.72
FedAvg	CIFAR-10	100	IID	20	37.4	482.9	0.69
FedAvg	CIFAR-10	100	Dir(0.01)	20	11.9	482.9	0.30
FedAvg	CIFAR-10	100	Dir(0.1)	20	33.4	482.9	0.62
FedAvg	CIFAR-10	100	Dir(0.5)	20	36.7	482.9	0.67
FedAvg	CIFAR-10	100	Dir(1.0)	20	36.7	482.9	0.67
FLwF-2	MNIST	10	IID	20	51.3	47.8	**0.00**
FLwF-2	MNIST	10	Dir(0.01)	20	9.8	47.8	0.15
FLwF-2	MNIST	10	Dir(0.1)	20	50.5	47.8	**0.01**
FLwF-2	MNIST	10	Dir(0.5)	20	51.2	47.8	**0.00**
FLwF-2	MNIST	10	Dir(1.0)	20	51.2	47.8	**0.00**
FLwF-2	MNIST	50	IID	20	51.1	239.1	**0.00**
FLwF-2	MNIST	50	Dir(0.01)	20	9.8	239.1	0.20
FLwF-2	MNIST	50	Dir(0.1)	20	50.6	239.1	**0.00**
FLwF-2	MNIST	50	Dir(0.5)	20	51.0	239.1	**0.00**
FLwF-2	MNIST	50	Dir(1.0)	20	51.1	239.1	**0.00**
FLwF-2	MNIST	100	IID	20	50.9	478.3	**0.00**
FLwF-2	MNIST	100	Dir(0.01)	20	30.0	478.3	**0.00**
FLwF-2	MNIST	100	Dir(0.1)	20	50.4	478.3	**0.00**
FLwF-2	MNIST	100	Dir(0.5)	20	50.8	478.3	**0.00**
FLwF-2	MNIST	100	Dir(1.0)	20	50.9	478.3	**0.00**
FLwF-2	FMNIST	10	IID	20	46.7	47.8	**0.00**

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